

1. **Elements that do not occur naturally and are made by scientists are called:**
 - A) Rare Earth elements
 - **B) Synthetic elements**
 - C) Noble elements
 - D) Transition elements
2. **Most synthetic elements are found in which part of the Periodic Table?**
 - A) Group 1
 - B) Group 17
 - **C) Actinide series**
 - D) Alkali metals
3. **The first element to be produced artificially was:**
 - **A) Technetium (Tc)**
 - B) Plutonium (Pu)
 - C) Uranium (U)
 - D) Neptunium (Np)
4. **Synthetic elements are generally created by bombarding heavy nuclei with:**
 - A) Water molecules
 - **B) Neutrons or Alpha particles**
 - C) Electrons only
 - D) Photons
5. **Which device is used to accelerate particles to high speeds to create new elements?**
 - A) Microscope
 - **B) Particle Accelerator (Cyclotron)**
 - C) Barometer
 - D) Galvanometer
6. **Elements with an atomic number greater than 92 (Uranium) are called:**
 - A) Sub-uranium elements
 - **B) Transuranic elements**
 - C) Isotopic elements
 - D) Lanthanides
7. **Technetium has the atomic number:**
 - A) 33

- ☐ **B) 43**
 - ☐ C) 53
 - ☐ D) 63
8. The symbol for the synthetic element "Americium" is:
- ☐ **A) Am**
 - ☐ B) Ar
 - ☐ C) Ac
 - ☐ D) Ai
9. Most man-made elements are highly:
- ☐ A) Stable
 - ☐ **B) Radioactive and Unstable**
 - ☐ C) Non-reactive
 - ☐ D) Liquid at room temperature
10. Which of the following is a man-made element used in smoke detectors?
- ☐ A) Gold
 - ☐ B) Iron
 - ☐ **C) Americium-241**
 - ☐ D) Carbon-12

Section 2: Transuranic Elements & Production

11. Neptunium (Np) was the first transuranic element discovered. Its atomic number is:
- ☐ A) 91
 - ☐ B) 92
 - ☐ **C) 93**
 - ☐ D) 94
12. The synthesis of elements usually involves a change in the:
- ☐ A) Number of valence electrons
 - ☐ **B) Atomic Number (Z)**
 - ☐ C) Physical state
 - ☐ D) Color
13. Plutonium (Pu) is primarily produced in:
- ☐ A) Chemical labs
 - ☐ **B) Nuclear reactors**
 - ☐ C) Volcanic eruptions

- D) The Earth's crust
- 14. **Curium (Cm) is named after which famous scientist(s)?**
 - A) Isaac Newton
 - **B) Marie and Pierre Curie**
 - C) Albert Einstein
 - D) John Dalton
- 15. **Which synthetic element is named after Albert Einstein?**
 - A) Albeium
 - **B) Einsteinium (Es)**
 - C) Fermium
 - D) Nobelium
- 16. **Elements like Mendelevium and Nobelium are part of the:**
 - **A) Actinides**
 - B) Halogens
 - C) Alkaline Earth metals
 - D) Metalloids
- 17. **The process of changing one element into another is called:**
 - A) Fusion
 - B) Fission
 - **C) Transmutation**
 - D) Evaporation
- 18. **Which particle is often emitted when a new element is formed via bombardment?**
 - **A) Neutron**
 - B) Molecule
 - C) Atom
 - D) Crystal
- 19. **Californium (Cf) is a synthetic element named after:**
 - A) A scientist
 - **B) A State/University in the USA**
 - C) A planet
 - D) A mineral
- 20. **The half-life of many laboratory-created elements is:**
 - A) Millions of years
 - B) Thousands of years
 - **C) Very short (seconds or minutes)**

- D) Infinite
- 21. **Promethium (Pm) is a rare synthetic element often used in:**
 - **A) Nuclear batteries (Luminous paint)**
 - B) Cooking salt
 - C) Fertilizers
 - D) Building houses
- 22. **Plutonium-239 is significant because it can undergo:**
 - A) Rusting
 - **B) Nuclear Fission**
 - C) Photosynthesis
 - D) Melting
- 23. **Which element is named after the "father of the periodic table"?**
 - A) Daltonium
 - **B) Mendelevium (Md)**
 - C) Seaborgium
 - D) Bohrsorium
- 24. **Laboratory-created elements are often produced in "traces." What does "trace" mean?**
 - A) Huge amounts
 - **B) Minute/Tiny amounts**
 - C) Liquid form
 - D) Gas form
- 25. **The study of creating new elements is part of:**
 - A) Organic Chemistry
 - B) Biology
 - **C) Nuclear Chemistry**
 - D) Geology
- 26. **Synthetic elements are placed after which element in the periodic table?**
 - A) Lead
 - B) Bismuth
 - **C) Uranium**
 - D) Radium
- 27. **The primary reason synthetic elements are not found in nature is because:**
 - A) They are too heavy to form.

- **B) Their half-lives are much shorter than the age of the Earth.**
 - C) They only exist as gases.
 - D) They are repelled by gravity.
28. **Which group of the periodic table contains the most synthetic elements?**
- A) Group 2 (Alkaline Earth Metals)
 - B) Group 18 (Noble Gases)
 - **C) The Actinide Series (Period 7)**
 - D) The Lanthanide Series (Period 6)
29. **Seaborgium (Sg) is named after Glenn T. Seaborg. Its atomic number is:**
- A) 100
 - **B) 106**
 - C) 110
 - D) 118
30. **When a nucleus is bombarded with a particle and becomes a new element, the process is called:**
- A) Ionization
 - **B) Induced Transmutation**
 - C) Distillation
 - D) Sublimation
31. **Which synthetic element is used as a power source for space probes like Voyager?**
- A) Technetium
 - **B) Plutonium-238**
 - C) Mendelevium
 - D) Einsteinium
32. **The "Island of Stability" is a theoretical concept suggesting that:**
- A) All synthetic elements will eventually become stable.
 - **B) Some very heavy synthetic elements might have longer half-lives.**
 - C) Synthetic elements can only be made on islands.
 - D) Protons and neutrons do not exist in heavy elements.
33. **Which of these is a synthetic element with atomic number 95?**
- A) Neptunium

- B) Plutonium
 - **C) Americium**
 - D) Curium
34. **Synthetic elements are typically:**
- **A) Metals**
 - B) Non-metals
 - C) Noble gases
 - D) Liquids
35. **The element Oganesson (Og) marks the end of:**
- A) Period 6
 - **B) Period 7**
 - C) Group 1
 - D) The transition metals
36. **Which particle was used to create the first synthetic element, Technetium?**
- **A) Deuterons (Heavy hydrogen nuclei)**
 - B) Electrons
 - C) Positrons
 - D) Photons
37. **The synthetic element used in portable X-ray machines is:**
- A) Uranium
 - B) Iron
 - **C) Californium-252**
 - D) Helium
38. **Fermium (Fm) was first discovered in:**
- A) A gold mine
 - **B) Debris from a hydrogen bomb explosion**
 - C) A volcanic crater
 - D) The atmosphere of Mars
39. **Most synthetic elements decay by emitting:**
- **A) Alpha or Beta particles**
 - B) Visible light
 - C) Radio waves
 - D) Sound waves
40. **The target material used to create new elements in a lab is often:**
- A) Plastic

- B) Wood
 - **C) A thin foil of a heavy element like Lead or Bismuth**
 - D) Glass
41. **Isotopes of synthetic elements have the same number of:**
- **A) Protons**
 - B) Neutrons
 - C) Mass units
 - D) Nucleons
42. **The atomic mass of synthetic elements in the periodic table is usually given as:**
- A) An average of all isotopes
 - **B) The mass number of the longest-lived isotope in brackets**
 - C) Zero
 - D) A whole number only
43. **Which element is used to produce Technetium-99m for medical imaging?**
- **A) Molybdenum-99**
 - B) Carbon-12
 - C) Oxygen-16
 - D) Nitrogen-14
44. **Bohrium (Bh) is named after:**
- A) John Dalton
 - **B) Niels Bohr**
 - C) Robert Boyle
 - D) Ernest Rutherford
45. **The creation of synthetic elements helps scientists understand:**
- A) How to make gold from lead easily
 - **B) The forces that hold the atomic nucleus together**
 - C) How to stop all radioactive decay
 - D) How to change the color of metals
46. **Which of the following is NOT a synthetic element?**
- A) Plutonium
 - B) Americium
 - **C) Uranium**
 - D) Curium

47. The element Dubnium (Db) is named after a research center in:
- A) USA
 - **B) Russia**
 - C) Germany
 - D) Pakistan
48. In the periodic table, the atomic number of synthetic elements:
- **A) Continues to increase**
 - B) Stays below 100
 - C) Is always an even number
 - D) Is always a prime number
49. Heavy ion accelerators are used to:
- A) Clean laboratory equipment
 - **B) Fuse two smaller nuclei into one larger synthetic nucleus**
 - C) Break atoms into pure energy
 - D) Cool the lab environment
50. The number of naturally occurring elements is approximately:
- A) 50
 - **B) 92**
 - C) 118
 - D) 200
51. The element named after the city of Berkeley, California, is:
- A) Californium
 - **B) Berkelium (Bk)**
 - C) Americium
 - D) Lawrencium
52. Which synthetic element is named after the inventor of the cyclotron, Ernest Lawrence?
- A) Ernestium
 - B) Cyclotronium
 - **C) Lawrencium (Lr)**
 - D) Nobelium
53. The element with atomic number 102 is named after Alfred Nobel. It is:

- A) Neptunium
 - B) Plutonium
 - **C) Nobelium (No)**
 - D) Hassium
54. **Which country has a synthetic element named "Nihonium" after its name in the local language?**
- A) China
 - **B) Japan**
 - C) Russia
 - D) Germany
55. **The temporary systematic names for undiscovered elements (like Ununseptium) are based on:**
- **A) Latin and Greek roots for their atomic numbers**
 - B) The name of the lead scientist
 - C) The laboratory building number
 - D) Randomly assigned letters
56. **Meitnerite (Mt) is named after Lise Meitner to honor her work in:**
- A) Photosynthesis
 - **B) Nuclear Fission**
 - C) Evolution
 - D) Acid-base reactions
57. **The element Livermorium (Lv) is named after a laboratory located in:**
- **A) California, USA**
 - B) London, UK
 - C) Moscow, Russia
 - D) Tokyo, Japan
58. **Which of these was the last element of the 7th period to be officially named?**
- A) Radon
 - B) Uranium
 - **C) Oganesson (Og)**
 - D) Radium
59. **Synthetic elements are often created using "Heavy Ion Fusion," which means:**
- A) Melting metal in a furnace

- **B) Fusing the nuclei of two heavy atoms**
 - C) Mixing chemicals in a test tube
 - D) Burning hydrogen gas
60. **Fermium-257 is notable because it is the heaviest isotope that can be formed by:**
- **A) Neutron capture**
 - B) Oxidation
 - C) Evaporation
 - D) Fermentation
61. **In a particle accelerator, magnets are used to:**
- A) Heat the atoms
 - **B) Steer and focus the beam of particles**
 - C) Change the color of the element
 - D) Measure the weight of the lab
62. **The "Target" in element synthesis is usually made of:**
- A) Plastic
 - B) Wood
 - **C) A heavy metal foil (like Bismuth or Curium)**
 - D) Liquid water
63. **To identify a single atom of a new element, scientists look at its:**
- **A) Decay chain (the particles it emits as it breaks down)**
 - B) Taste and smell
 - C) Reaction with vinegar
 - D) Boiling point
64. **Many synthetic elements exist for only "milliseconds." A millisecond is:**
- A) 1000 seconds
 - B) 60 seconds
 - **C) 1/1000th of a second**
 - D) 1/10th of a second
65. **The IUPAC is the organization that:**
- A) Builds nuclear reactors
 - **B) Approves the names of new elements**
 - C) Sells synthetic elements
 - D) Mines Uranium

66. **As atomic number increases in synthetic elements, the nucleus usually becomes:**
- A) More stable
 - **B) Less stable (shorter half-life)**
 - C) Smaller in size
 - D) Non-radioactive
67. **Which synthetic element is used to treat certain types of brain tumors?**
- A) Iron
 - B) Sodium
 - **C) Californium-252 (as a neutron source)**
 - D) Gold
68. **The "Superheavy" elements are those with atomic numbers starting from:**
- A) 10
 - B) 50
 - **C) 104**
 - D) 92
69. **Synthetic elements are typically prepared in:**
- A) Open fields
 - **B) High-vacuum chambers**
 - C) Underwater labs
 - D) Regular school classrooms
70. **Roentgenium (Rg) is named after the discoverer of:**
- A) Gravity
 - **B) X-rays**
 - C) Electricity
 - D) The Microscope
71. **Which element is a synthetic member of the Noble Gas group? (Ans: Oganesson)**
72. **What is the atomic number of Plutonium? (Ans: 94)**
73. **Can synthetic elements be seen with the naked eye? (Ans: Rarely, most are produced atom-by-atom)**
74. **Is Lead a synthetic element? (Ans: No, it is natural)**
75. **What is the primary fuel for a breeder reactor to make Plutonium? (Ans: Uranium-238)**
76. **Which radiation is a Helium nucleus? (Ans: Alpha)**

77. Which synthetic element is used in industrial thickness gauges? (Ans: Americium-241)
78. What happens to the mass number in Beta decay? (Ans: Remains the same)
79. What happens to the atomic number in Alpha decay? (Ans: Decreases by 2)
80. Who is the only living person to have an element named after them at the time? (Ans: Seaborg/Oganesson)
81. What is the symbol for Hassium? (Ans: Hs)
82. Are all actinides radioactive? (Ans: Yes)
83. Which element is used to calibrate radiation counters? (Ans: Technetium-99)
84. What is the atomic number of Mendelevium? (Ans: 101)
85. What is the charge of a Beta particle? (Ans: -1)
86. Where is the Dubna Joint Institute for Nuclear Research located? (Ans: Russia)
87. Which synthetic element is named after Germany? (Ans: Darmstadtium/Hassium references)
88. What is the heaviest naturally occurring element? (Ans: Uranium)
89. What is the symbol for Rutherfordium? (Ans: Rf)
90. Is Einsteinium natural or synthetic? (Ans: Synthetic)
91. What is the atomic number of Neptunium? (Ans: 93)
92. Which synthetic element is used in some heart pacemakers? (Ans: Plutonium-238)
93. What is the symbol for Flerovium? (Ans: Fl)
94. Does Gamma decay change the atomic number? (Ans: No)
95. What is the atomic number of Lawrencium? (Ans: 103)
96. Which lab discovered Nihonium? (Ans: RIKEN in Japan)
97. What is the symbol for Moscovium? (Ans: Mc)
98. Is Tennessine a metal or halogen-like? (Ans: Halogen-like)
99. What is the atomic number of Oganesson? (Ans: 118)
100. Why are synthetic elements important? (Ans: They expand our knowledge of nuclear physics and chemistry.)